

WCMS900B170E53

1700V All-SiC Power Module Half-Bridge Module

$V_{DS}=1700V$

$R_{DS}=1.5m\Omega$

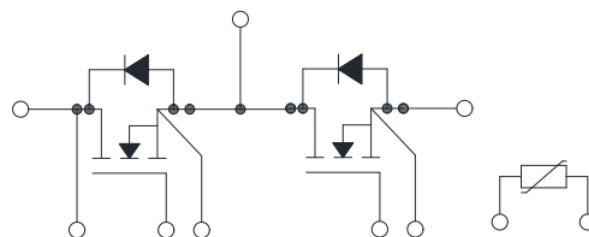
Features

- Ultra Low Loss
- High-Frequency Operation
- Zero Reverse Recovery Current from Diode
- Zero Turn-Off Tail Current from MOSFET
- Ease of Paralleling



Typical Applications

- Induction Heating
- Motor Drives
- Renewables
- Railway Auxiliary & Traction
- EV Fast Charging
- UPS and SMPS



1700V/1.5mΩ in one-package

● Absolute Maximum Ratings ($T_{vj}=25^{\circ}C$ unless otherwise specified)

Parameter	Symbol	Conditions	Value	Unit
Drain-Source Voltage	$V_{DS(max)}$		1700	V
Continuous Drain Current	I_D	$V_{GS}=18V, T_c=25^{\circ}C$	-	A
		$V_{GS}=18V, T_c=80^{\circ}C$	-	
Gate- Source Voltage	$V_{GS(max)}$	Absolute Maximum Values	-10V/+23V	V
Gate-Source Voltage	$V_{GS(op)}$	Recommended Operational Values	-5V/+18V	V
Operating Junction Temperature	$T_{vj(op)}$		-40~+150	$^{\circ}C$
Maximum Junction Temperature	$T_{vj(max)}$		175	$^{\circ}C$
Total power dissipation	P_{tot}	$T_c=80^{\circ}C$	-	W

● MOSFET Characteristics ($T_{vj}=25^{\circ}\text{C}$ unless otherwise specified)

Parameter	Symbol	Conditions	Value			Unit
			Min.	Typ.	Max.	
Gate Threshold Voltage	$V_{GS(th)}$	$V_{DS}=V_{GS}, I_D=360\text{mA}$	2.0	3.2	4.0	V
		$V_{DS}=V_{GS}, I_D=360\text{mA}, T_{vj}=175^{\circ}\text{C}$		2.2		V
Zero Gate Voltage Drain Current	I_{DSS}	$V_{DS}=1700\text{V}, V_{GS}=0\text{V}$		10	120	μA
Gate-Source Leakage Current	I_{GSS}	$V_{GS}=18\text{V}$		100	1200	nA
On State Resistance	$R_{DS(on)}$	$V_{GS}=18\text{V}, I_{DS}=900\text{A}$		1.5	1.92	m Ω
		$V_{GS}=18\text{V}, I_{DS}=900\text{A}, T_{vj}=175^{\circ}\text{C}$		2.67		m Ω
Input Capacitance	C_{iss}	$V_{GS}=0\text{V}, V_{DS}=1000\text{V}, V_{AC}=25\text{mV}$ $f=100\text{kHz}$		82.35		nF
Output Capacitance	C_{oss}			2.4		nF
Reverse Transfer Capacitance	C_{rss}			0.3		nF
Gate-Source Charge	Q_{GS}	$V_{DS}=1000\text{V}, V_{GS}=-5\text{V}/+18\text{V}$ $I_D=900\text{A}$		936		nC
Gate-Drain Charge	Q_{GD}			600		nC
Total Gate Charge	Q_G			4344		nC
Turn-On Delay Time	$t_{d(on)}$	$I_D=900\text{A}$ $V_{DS}=800\text{V}$ $V_{GS}=+18\text{V}/-5\text{V}$ $R_G=2.5\Omega$ Inductive Load $T_{vj}=25^{\circ}\text{C}$		197		ns
Rise Time	t_r			115		ns
Turn-Off Delay Time	$t_{d(off)}$			842		ns
Fall Time	t_f			92		ns
Energy Dissipation during Turn-On Time	E_{on}			57.34		mJ
Energy Dissipation during Turn-Off Time	E_{off}	$I_D=900\text{A}$ $V_{DS}=800\text{V}$ $V_{GS}=+18\text{V}/-5\text{V}$ $R_G=5.1\Omega$ Inductive Load $T_{vj}=150^{\circ}\text{C}$		103.27		mJ
Turn-On Delay Time	$t_{d(on)}$			168		ns
Rise Time	t_r			155		ns
Turn-Off Delay Time	$t_{d(off)}$			1077		ns
Fall Time	t_f			57.6		ns
Energy Dissipation during Turn-On Time	E_{on}			97.2		mJ
Energy Dissipation during Turn-Off Time	E_{off}			41.3		mJ

● Body Diode Characteristics

Parameter	Symbol	Conditions	Value	Unit
DC Body Diode Forward Current	I_{SD}	$V_{GS}=-5V, T_c=80^{\circ}C$	-	A
Pulsed body diode current	$I_{SD \text{ pulse}}$	verified by design, t_p limited by T_{vjmax}	-	A

Parameter	Symbol	Conditions	Value			Unit
			Min.	Typ.	Max.	
Diode Forward Voltage	V_{SD}	$I_F=900A,$ $V_{DS}=800V$ $R_G=5.1\Omega$ $V_{GS}=-5V$	$T_{vj}=25^{\circ}C$	4.2		V
			$T_{vj}=175^{\circ}C$	3.8		
Forward Recovery Time	t_{rr}		$T_{vj}=25^{\circ}C$	49		ns
			$T_{vj}=150^{\circ}C$	72		
Peak Reverse Recovery Current	I_{rrm}		$T_{vj}=25^{\circ}C$	200.7		A
			$T_{vj}=150^{\circ}C$	243.2		
Recovered Charge	Q_{rr}		$T_{vj}=25^{\circ}C$	5.9		uC
			$T_{vj}=150^{\circ}C$	8.4		
Reverse Recovery Energy	E_{rec}		$T_{vj}=25^{\circ}C$	1.33		mJ
			$T_{vj}=150^{\circ}C$	8.58		

● NTC Thermistor

Parameter	Symbol	Conditions	Value			Unit
			Min.	Typ.	Max.	
Rated Resistance	R_{25}			5.0		kΩ
Deviation of R100	$\Delta R/R$	$T_C=100, R_{100}=493.3\Omega$	-5		5	%
Power Dissipation	P_{25}			20.0		mW
B-Value	$B_{25/50}$	$R_2=R_{25}\exp[B_{25/50}(1/T_2-1/(298.15K))]$		3375		K

● Module Characteristics

Parameter	Symbol	Conditions	Value			Unit
			Min.	Typ.	Max.	
Case Isolation Voltage	V_{isol}	$t=1min, f=50Hz$	2500			V
Maximum Junction Temperature	$T_{vj(max)}$				175	°C
Operating Junction Temperature	$T_{vj(op)}$		-40		150	°C
Storage Temperature	T_{stg}		-40		125	°C
Thermal Resistance Junction to Case	R_{thJC}	Per MOSFET		-	-	K/W
Module Electrodes Torque	M_t	Recommended(M6)	3.0		6.0	Nm
Module to Heatsink Torque	M_s	Recommended(M5)	3.0		6.0	
Weight of Module	G			346		g

● Typical Performance

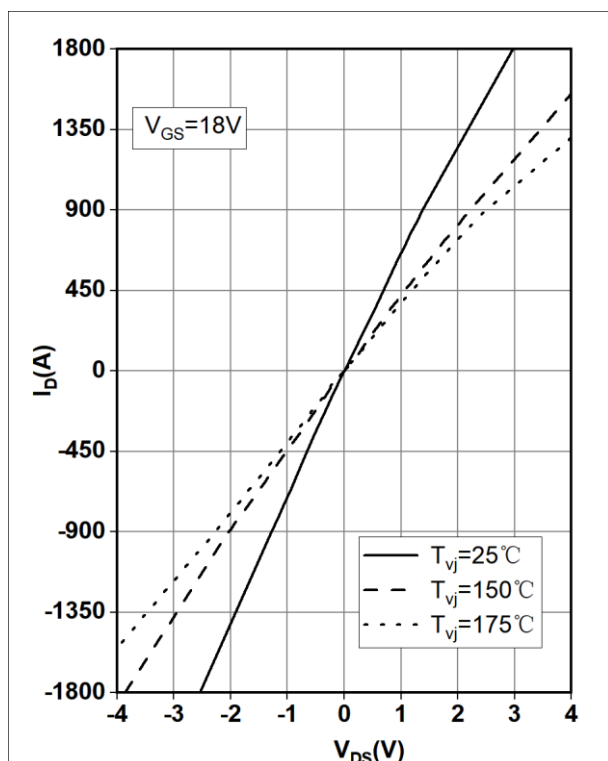


Fig.1 Output Characteristic (MOSFET), $I_D=f(V_{DS})$

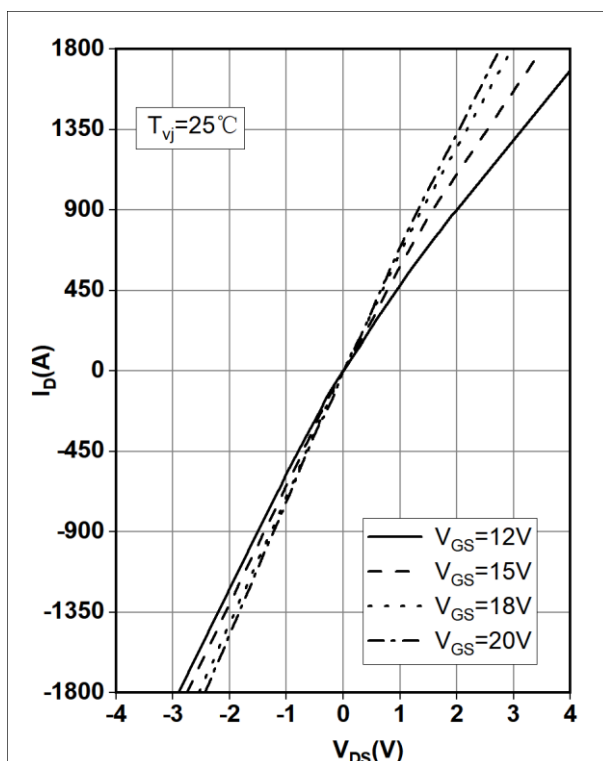


Fig.2 Output Characteristic (MOSFET), $I_D=f(V_{DS})$

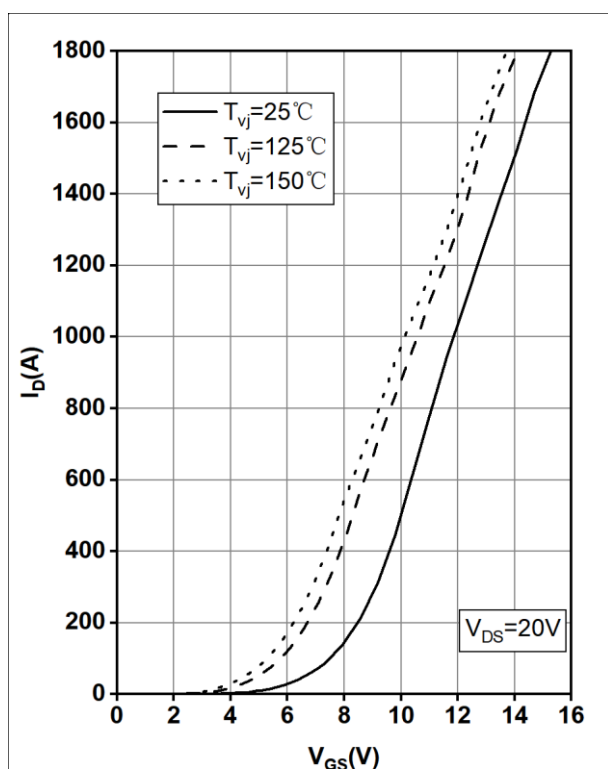


Fig.3 Transfer Characteristic (MOSFET), $I_D=f(V_{GS})$

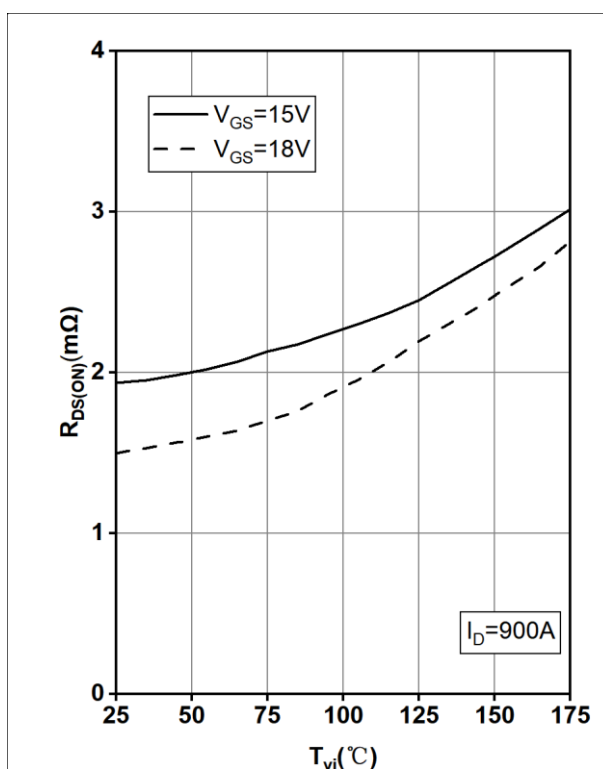


Fig.4 Drain Source On-resistance (MOSFET), $R_{DS(on)}=f(T_{vj})$

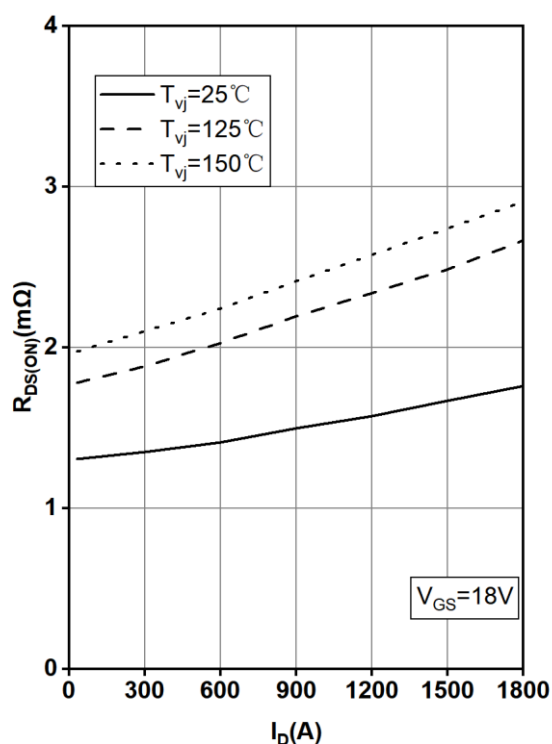


Fig.5 Drain Source On-resistance (MOSFET),
 $R_{DS(on)}=f(I_D)$

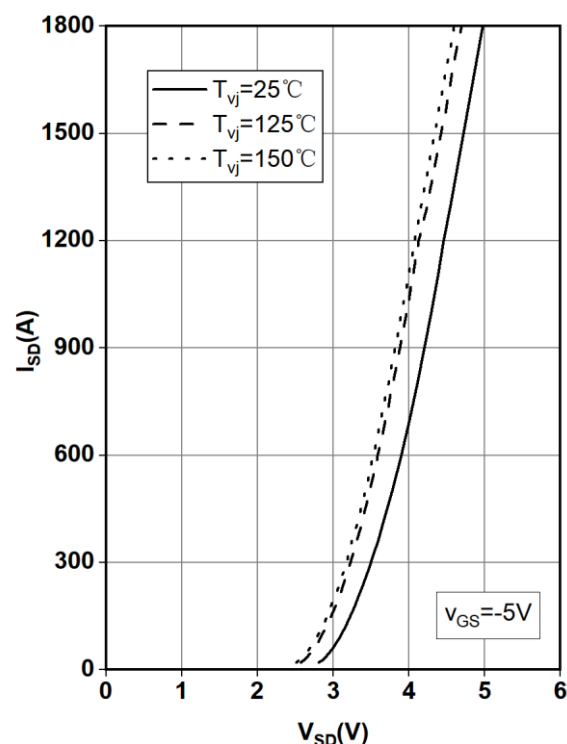


Fig.6 Body Diode Forward Characteristic (MOSFET),
 $V_{SD}=f(I_{SD})$

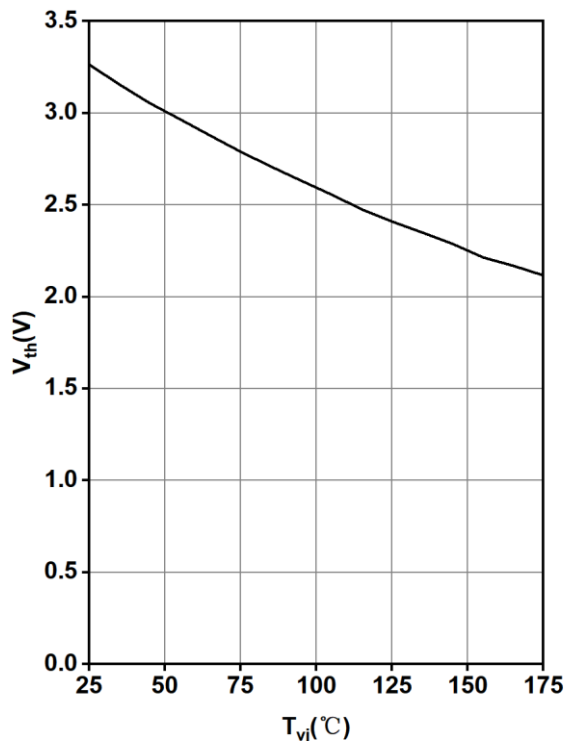


Fig.7 Voltage-Gate threshold (MOSFET), $V_{th}=f(T_{vj})$

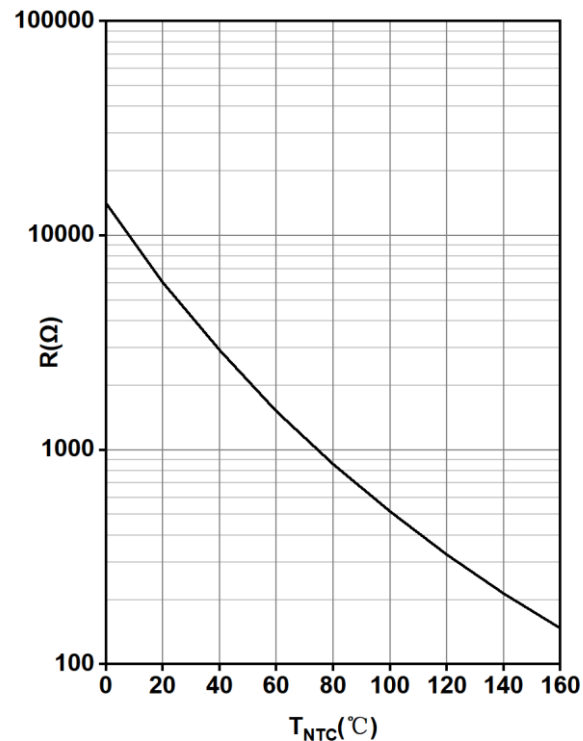


Fig.8 Temperature Characteristic (NTC-Thermistor),
 $R=f(T_{NTC})$

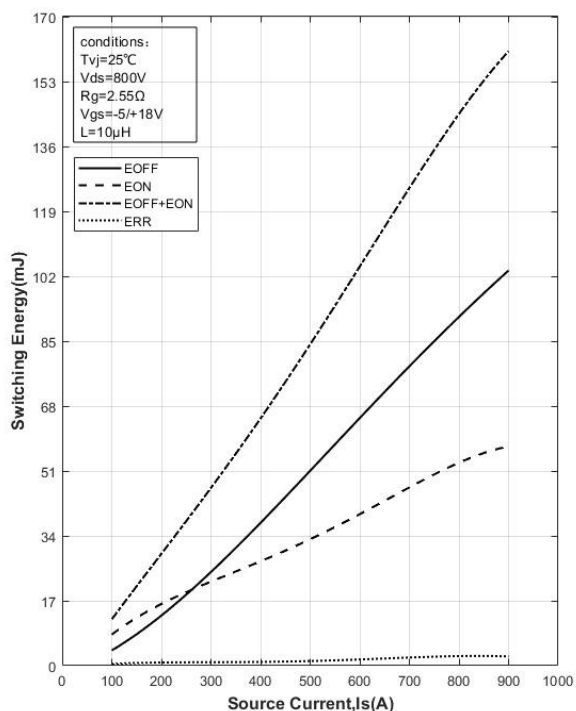


Fig 9 Switching Energy vs. Drain Current ($V_{ds} = 800V$)

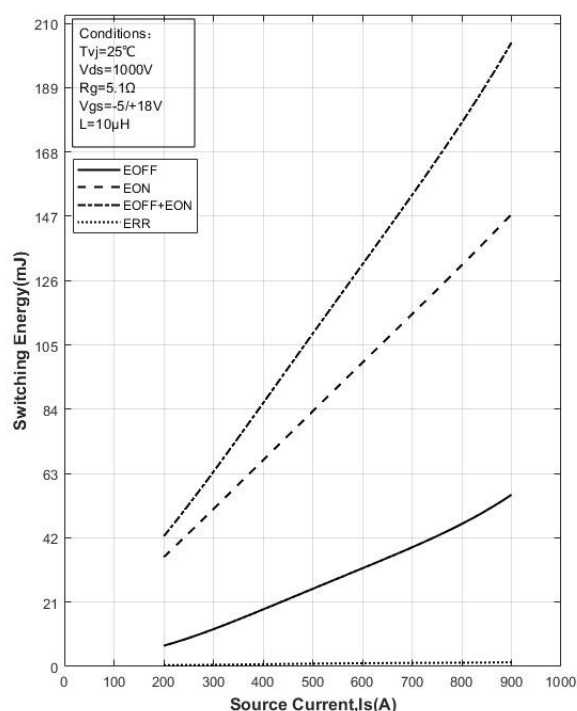


Fig 10 Switching Energy vs. Drain Current ($V_{ds} = 1000V$)

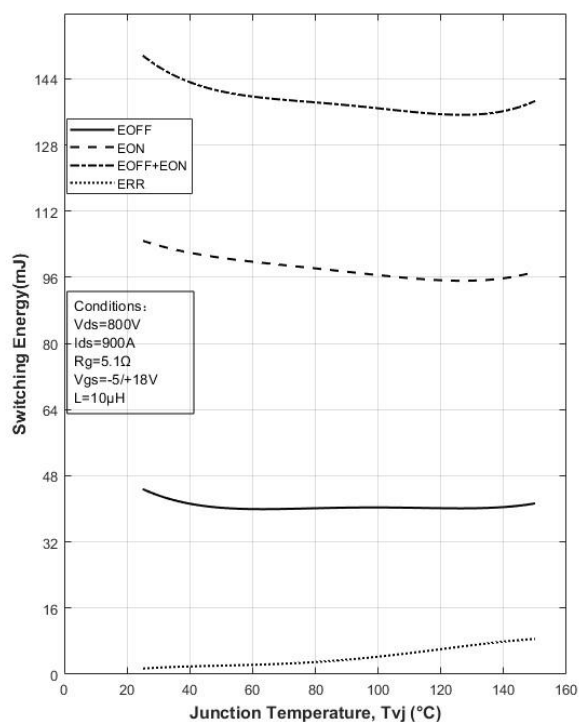


Fig 11 MOSFET Switching Energy vs. Junction Temperature

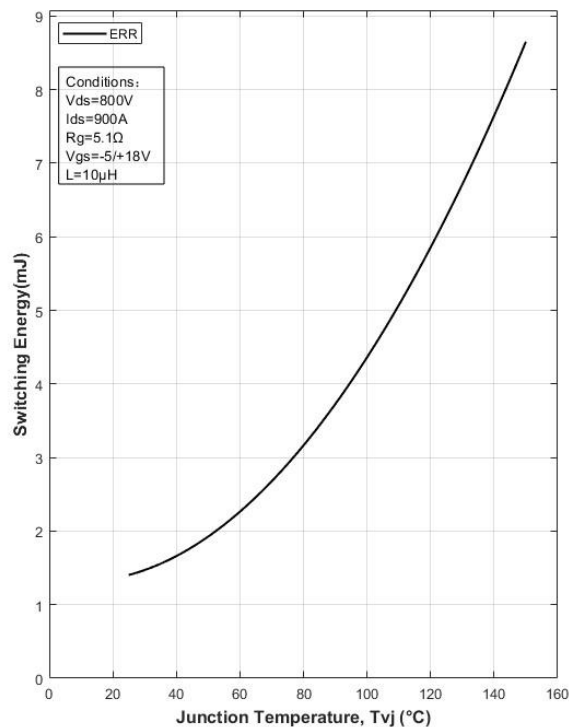


Fig 12 Reverse Recovery Energy vs. Junction Temperature

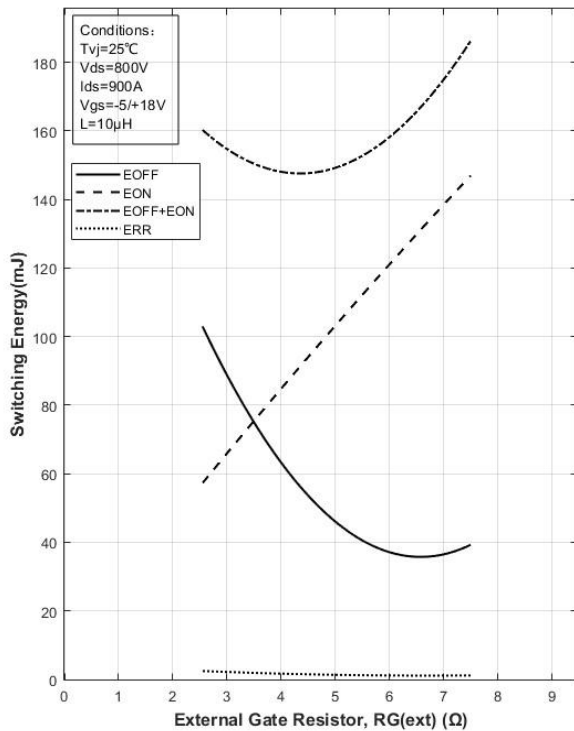


Fig 13 MOSFET Switching Energy vs. External GateResistance

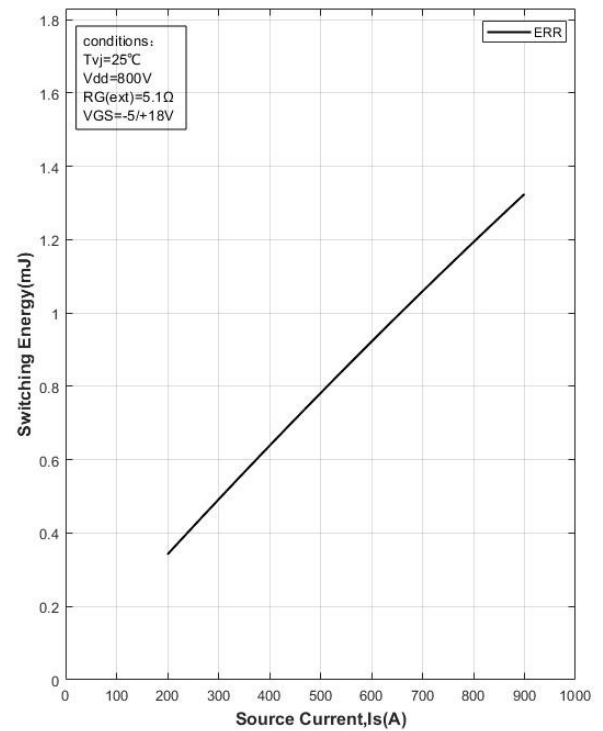


Fig 14 Reverse Recovery Energy vs. Source Current

● Timing Characteristics

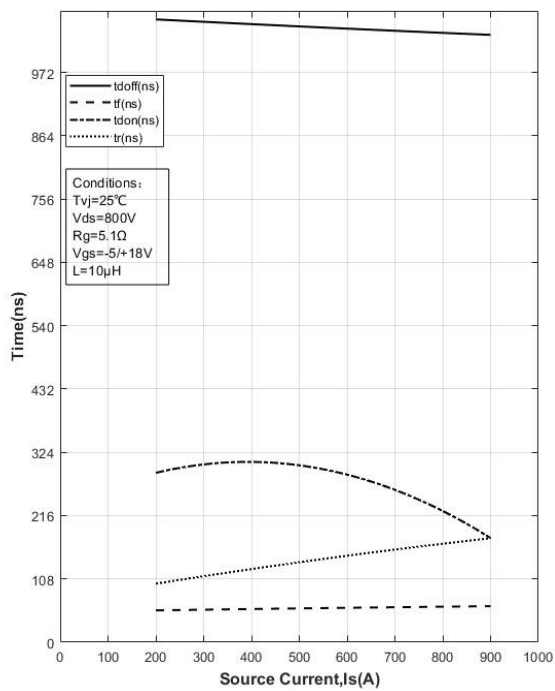


Fig 15 Timing vs. Source Current

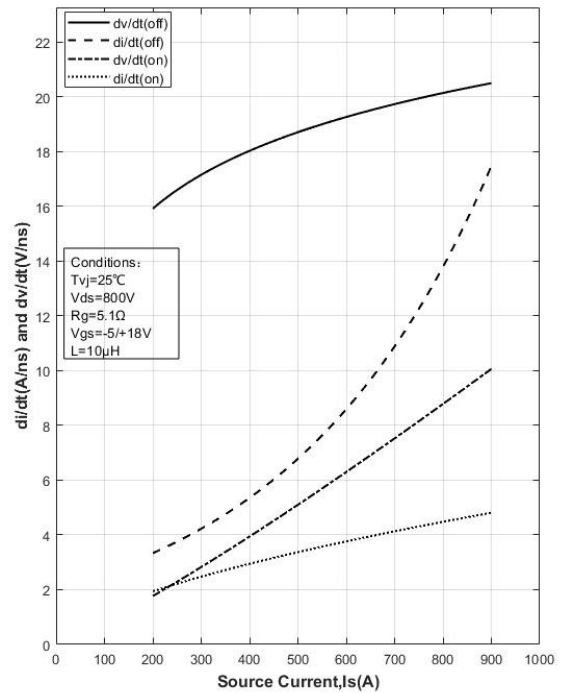


Fig 16 dv/dt and di/dt vs. Source Current

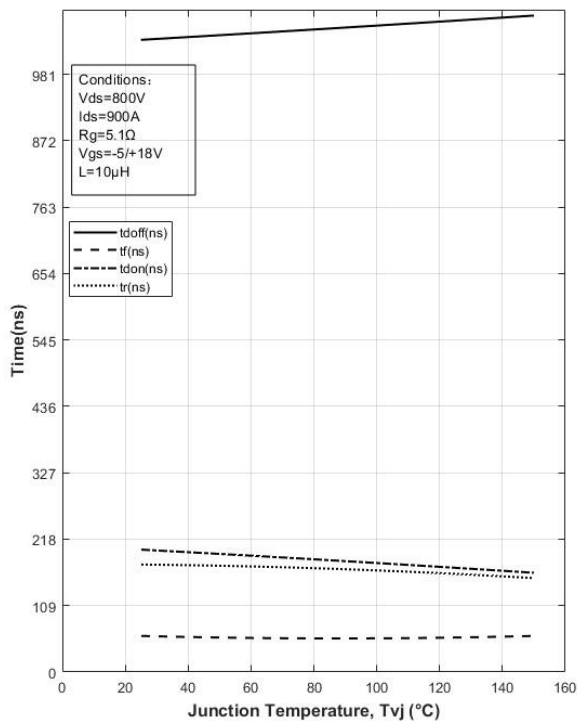


Fig 17 Timing vs. Junction Temperature

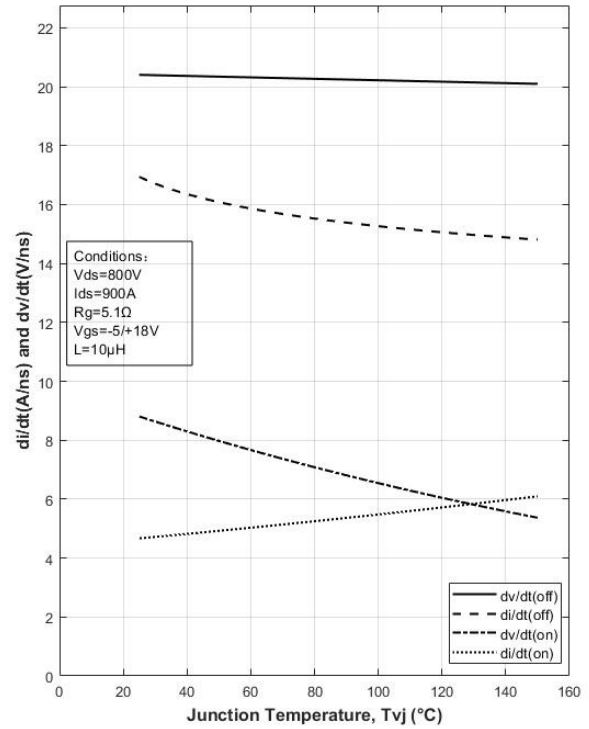


Fig 18 dv/dt and di/dt vs. Junction Temperature

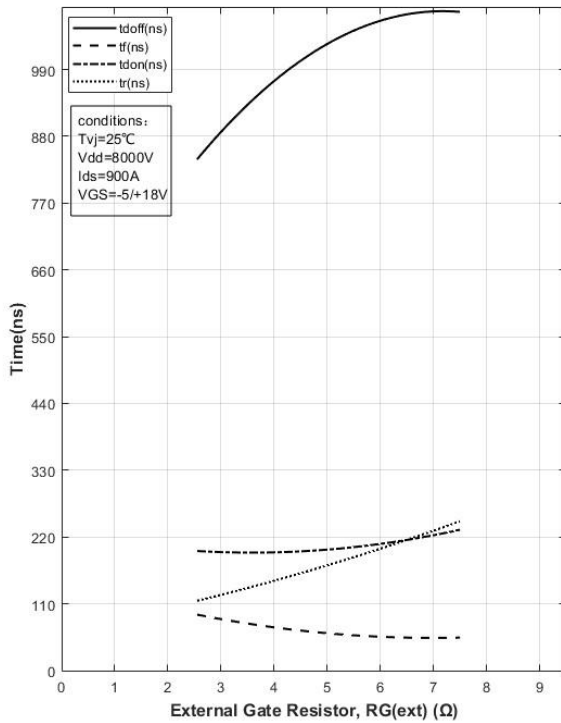


Fig 19 Timing vs. External Gate Resistance

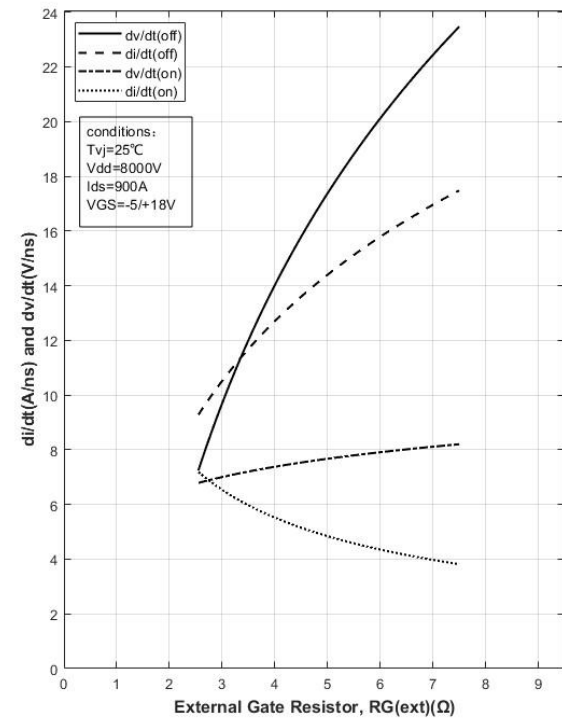
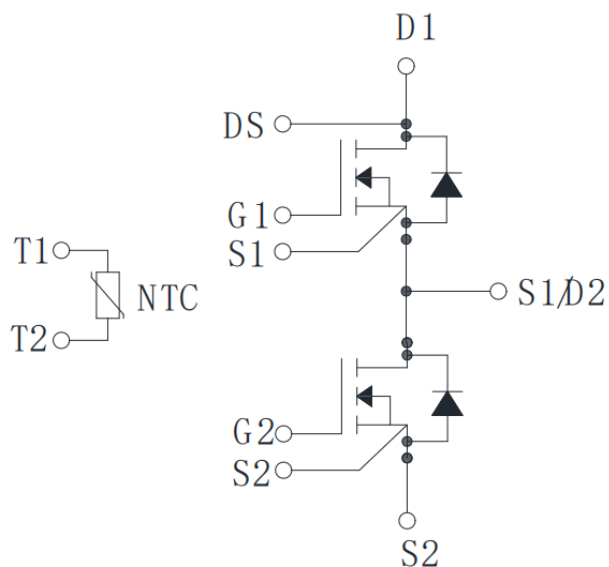
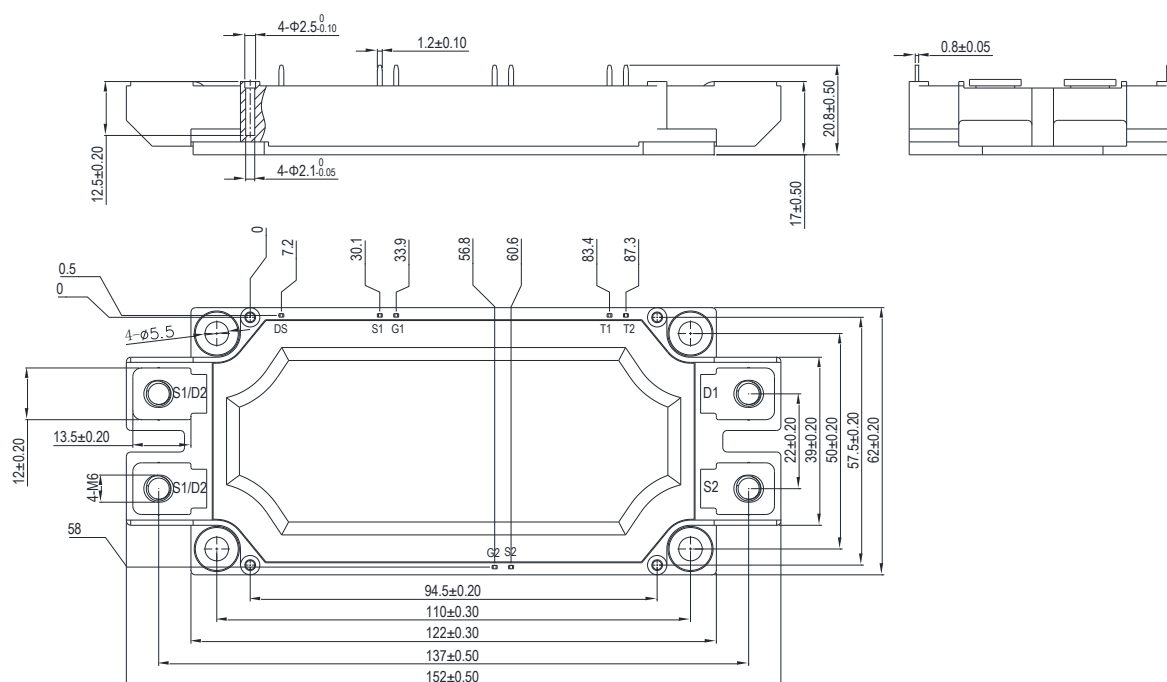


Fig 20 Timing vs. External Gate Resistance

● Circuit Diagram



● Package Dimensions



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